

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE CODE: MLA03**

**COURSE NAME: Reinforcement learning**

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### COURSE OUTCOME COVERED

***CO5:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

*Assessment Tool Weightage: 35%*

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**Dr G .A. SENTHIL**

**QUESTIONS: 10**

**Total Marks: 50**

Annexure A

**Scenario-Based Assessment**

**DURATION: 1.5 Hrs**

1. A mobile app company wants to improve user engagement by showing personalized notifications at the right time. The company plans to use Reinforcement Learning to learn user behavior patterns. Report how the RL framework can be applied by defining the possible states, actions, and reward functions for this scenario.

### **RL Framework for Personalized Mobile Notifications**

#### **1. Introduction**

Reinforcement Learning can be used to learn **when and what type of notification a user is most likely to engage with**. The mobile app acts as the environment, while the RL agent learns from the user's responses to notifications.

The main components are:

**State → Action → Reward → Next State → Policy Update**

#### **2. State Space**

The **state** represents the current information about the user.

Possible state variables include:

- User's current time of day
- Day of the week
- Recent app usage
- Number of recent sessions
- User's previous notification responses
- Type of content previously viewed
- Time since last app usage
- User's engagement level

### Example

```
State = {
  Time = 7 PM,
  Recent Usage = High,
  Last Notification = Opened,
  Preferred Content = Sports,
  Sessions Today = 4
}
```

### 3. Action Space

The RL agent decides what action to take.

Possible actions:

1. Send notification immediately
2. Delay notification
3. Do not send notification
4. Send promotional notification
5. Send personalized content notification
6. Send reminder notification

For example:

Action = Send personalized notification at 7:30 PM

### 4. Reward Function

The reward tells the agent whether its decision was good or bad.

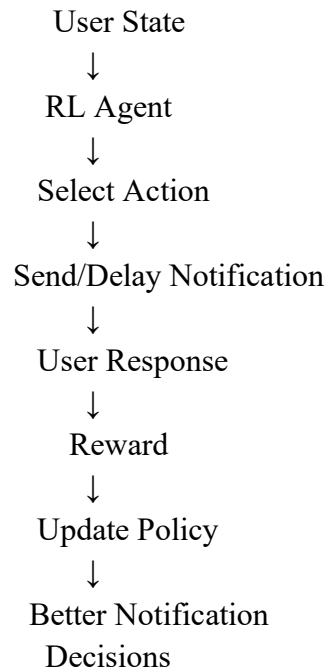
Possible rewards:

| User response               | Reward |
|-----------------------------|--------|
| Opens notification          | +5     |
| Uses app after notification | +10    |
| Makes desired interaction   | +15    |
| Ignores notification        | 0      |
| Dismisses notification      | -2     |
| Disables notifications      | -10    |

A simple reward function could be:

The reward should also penalize excessive notifications so that the system does not simply maximize notification frequency.

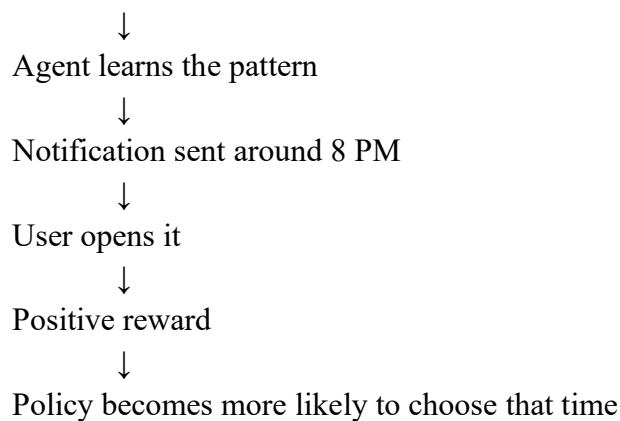
## 5. RL Working



The agent continuously observes user behavior and updates its policy.

For example:

User usually opens notifications at 8 PM



## 6. Objective

The goal is to maximize **long-term user engagement** while minimizing:

- Notification fatigue
- Irrelevant notifications
- Excessive notification frequency
- User opt-outs

The objective can be represented as:

where:

- = notification policy
- = reward at time
- = discount factor

## 7. Example

Suppose a user normally uses the app between **7 PM and 9 PM**.

The agent tries three actions:

6 PM → Notification → Ignored → Reward 0

7:30 PM → Notification → Opened → Reward +5

10 PM → Notification → Dismissed → Reward -2

After repeated interactions, the RL agent learns that **7:30 PM** produces better engagement and adjusts its policy accordingly.

## 8. Conclusion

Thus, the mobile app can be modeled as an RL environment where **user behavior forms the state, notification decisions form the actions, and user engagement forms the reward**. By continuously learning from feedback, the RL agent can discover personalized notification timing and content while reducing unnecessary notifications.

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### **Code of Conduct:**

I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

### **RUBRICS FOR CALCULATION**

| Criteria                  | Weightage (%) | Level 4 (Exemplary)                                                       | Level 3 (Proficient)                     | Level 2 (Developing)                       | Level 1 (Beginning)                               |
|---------------------------|---------------|---------------------------------------------------------------------------|------------------------------------------|--------------------------------------------|---------------------------------------------------|
| Understanding of Scenario | 20%           | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding ; misses key aspects | Misinterprets or fails to understand the scenario |

|                              |     |                                                                  |                                                    |                                                   |                                               |
|------------------------------|-----|------------------------------------------------------------------|----------------------------------------------------|---------------------------------------------------|-----------------------------------------------|
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved   | Identifies most key issues with minor omissions    | Identifies limited issues; some irrelevant points | Unable to identify key issues                 |
| Application of Concepts      | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors                 | Limited or partially correct application          | Unable to apply relevant concepts             |
| Decision Making              | 20% | Makes well-reasoned, logical decisions with strong rationale     | Makes reasonable decisions with some justification | Makes weak decisions with limited justification   | No clear decisions made or rationale provided |
| Clarity of Response          | 15% | Response is clear, structured, and logically presented           | Generally clear with minor issues                  | Somewhat unclear or poorly structured             | Disorganized and difficult to understand      |